

# M4\_Assignment

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## Q2 Home Prices

### Q2.1 Loading packages and data

Write code below that loads fpp3, tidyverse packages. Then load HomePrices.csv data set with the read\_csv() function and save it in an object called Homes. Use the glimpse() function to preview the data. Include the code below:

```
library(tidyverse)
library(fpp3)

Homes <- read_csv("https://jagelives.github.io/Data/HomePrices.csv")

glimpse(Homes)
```

Rows: 347

Columns: 4

```
$ saledate <chr> "30/09/2007", "31/12/2007", "31/03/2008", "30/06/2008", "30/0~
$ Price    <dbl> 441854, 441854, 441854, 441854, 451583, 440256, 442566, 44611~
$ type     <chr> "house", "house", "house", "house", "house", "house", "house"~
$ bedrooms <dbl> 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2~
```

### Q2.3 Piping

Start with the Homes object. Use the pipe operator %>% to chain the following commands.

- filter the data so that only “two-bedroom” houses are included
- coerce the saledate variable to a date using the dmy() function. format the saledate variable to quarters by using the yearquarter() function.

- remove the bedrooms and type variables. You should include a pipe for each bullet point and save the result to a new object called h2.

```
Homes %>%
  filter(bedrooms == 2, type == "house") %>%
  mutate(saledate = yearquarter(dmy(saledate))) %>%
  select(saledate, Price) -> h2

head(h2)
```

```
# A tibble: 6 x 2
  saledate Price
  <qtr>    <dbl>
1 2007 Q3 441854
2 2007 Q4 441854
3 2008 Q1 441854
4 2008 Q2 441854
5 2008 Q3 451583
6 2008 Q4 440256
```

## Q2.4 tsibble

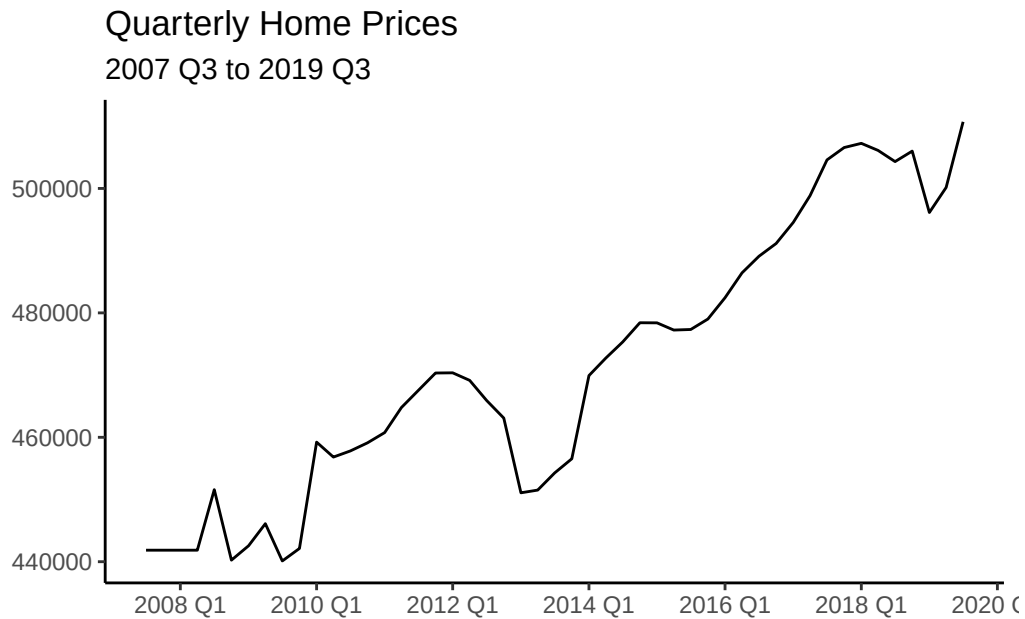
Coerce the h2 tibble to a tsibble using the `as_tsibble()` function. Make sure you specify the correct index. Save the new tsibble in an object called h2\_ts. Include your code below:

```
h2_ts <- as_tsibble(h2, index=saledate)
```

## Q2.5 Plot

Create a line plot of the Price variable using the `ggplot()` function. Add a title that reads “Quarterly Home Prices” and a subtitle that reads “2007 Q3 to 2019 Q3”. Omit the x and y captions.

```
ggplot(data=h2_ts) +
  geom_line(mapping = aes(x=saledate, y=Price), color='black') +
  theme_classic() +
  labs(x="",
       y="",
       title="Quarterly Home Prices",
       subtitle="2007 Q3 to 2019 Q3")
```



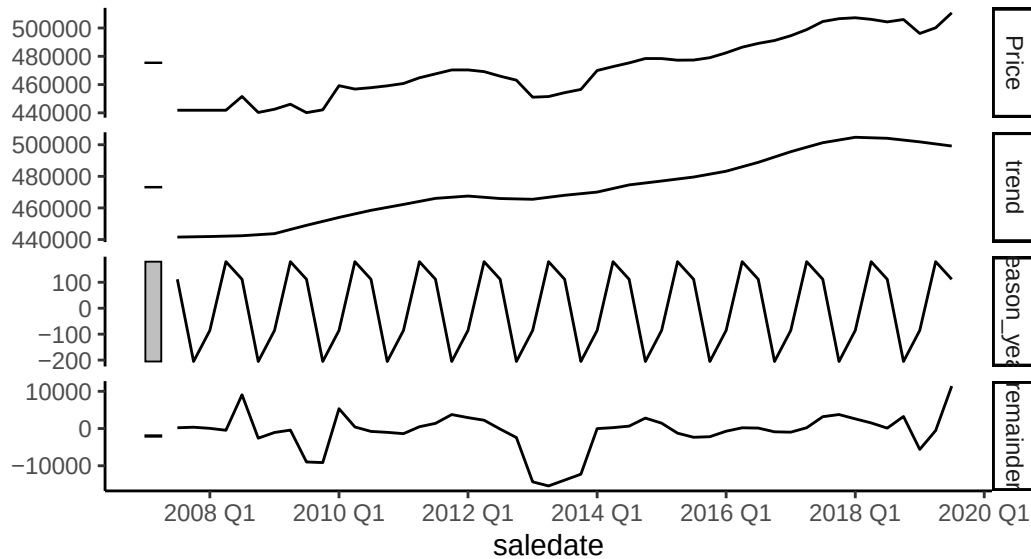
## Q2.6 STL

Perform an STL decomposition using the `model()` and `STL()` functions. Set the window argument to 11 for the `trend()`. Set 'periodic' for the window argument of the `season()` function.

```
h2_ts %>%
  model(STL(Price~trend(window=11)+season(window='periodic'), robust=TRUE)) %>%
  components() %>% autoplot() + theme_classic()
```

## STL decomposition

Price = trend + season\_year + remainder



### Q2.7 STL Components

```
h2_ts %>%
  model(STL(Price~trend(window=11)+season(window='periodic'), robust=TRUE)) %>%
  components() -> h2_components
h2_components
```

# A dable: 49 x 7 [1Q]

# Key: .model [1]

# : Price = trend + season\_year + remainder

|   | .model             | saledate | Price  | trend  | season_year | remainder | season_adjust |
|---|--------------------|----------|--------|--------|-------------|-----------|---------------|
|   | <chr>              | <qtr>    | <dbl>  | <dbl>  | <dbl>       | <dbl>     | <dbl>         |
| 1 | "STL(Price ~ tren~ | 2007 Q3  | 441854 | 4.42e5 | 112.        | 209.      | 441742.       |
| 2 | "STL(Price ~ tren~ | 2007 Q4  | 441854 | 4.42e5 | -205.       | 350.      | 442059.       |
| 3 | "STL(Price ~ tren~ | 2008 Q1  | 441854 | 4.42e5 | -85.5       | 53.8      | 441940.       |
| 4 | "STL(Price ~ tren~ | 2008 Q2  | 441854 | 4.42e5 | 179.        | -468.     | 441675.       |
| 5 | "STL(Price ~ tren~ | 2008 Q3  | 451583 | 4.42e5 | 112.        | 9073.     | 451471.       |
| 6 | "STL(Price ~ tren~ | 2008 Q4  | 440256 | 4.43e5 | -205.       | -2595.    | 440461.       |
| 7 | "STL(Price ~ tren~ | 2009 Q1  | 442566 | 4.44e5 | -85.5       | -1062.    | 442652.       |
| 8 | "STL(Price ~ tren~ | 2009 Q2  | 446113 | 4.46e5 | 179.        | -418.     | 445934.       |
| 9 | "STL(Price ~ tren~ | 2009 Q3  | 440123 | 4.49e5 | 112.        | -8979.    | 440011.       |

```
10 "STL(Price ~ tren~ 2009 Q4 442131 4.51e5 -205. -9147. 442336.  
# i 39 more rows
```